$$\phi_t = k\,|w_t(1-\phi_t) - d_t| = \begin{cases} \dfrac{k(w_t - d_t)}{1 + kw_t}, & w_t \geq d_t, \\ \dfrac{k(d_t - w_t)}{1 - kw_t}, & w_t < d_t. \end{cases}$$